

UGV01 Live Contract Experiment Guide

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This guide explains how to run the **live contract experiment** with the current repository and UGV01 workflow

Short answer:

For the live contract dashboard, keep **GPS connected**.

You do **not** need AprilTags for the live demo itself.

AprilTags are only needed later for independent offline validation metrics.

1. What This Experiment Is

The live contract experiment is the online digital-twin dashboard demonstration in which the rover is controlled from the interface, the twin is updated from onboard telemetry, service contracts are evaluated online, and the resource policy chooses 2 Hz, 5 Hz, or 10 Hz behavior.

```
python -m DigitalTwin.dashboard.server --mode live --rover-url http://192.168.4.1/js --host 127.0.0.1 --p
```

2. What Must Be Connected

Required for the live contract experiment

- UGV01 rover powered on
- Wi-Fi connection to the rover or its station-mode IP
- Firmware streaming telemetry at /js
- Working IMU
- Working encoders

Strongly recommended

- **GPS connected and functioning**

The current live dashboard uses GPS as the live operational reference for contract-style monitoring. Without GPS, the dashboard can still run, but the live fidelity and contract evidence becomes much weaker.

Not required for the live contract experiment

- AprilTags
- ChArUco board
- Overhead phone video

Experiment	GPS	AprilTags
Live contract dashboard	Yes, recommended	No
Offline physical validation	Optional	Yes

3. Rover Bring-Up

Active firmware path: ugv01_gps_dev/General_Driver

BN220 wiring:

```
BN-220 white -> UGV01 RX
BN-220 red   -> UGV01 5V
BN-220 black -> UGV01 GND
```

- 1 Power on the rover.
- 2 Connect your laptop to the rover Wi-Fi or station-mode Wi-Fi.
- 3 Open `http://192.168.4.1`.
- 4 Confirm the stock control page responds and telemetry updates.
- 5 Confirm GPS values update if GPS is attached.

If the rover is on station-mode Wi-Fi, replace `192.168.4.1` with the IP shown on the rover display.

4. Recommended Experiment Order

Step 1: Rehearse the dashboard without the rover

```
python -m DigitalTwin.dashboard.server --mode csv --csv raw_logs\telemetry\ugv_t147_bench_20260814_143729
```

Step 2: Run the actual live rover dashboard

```
python -m DigitalTwin.dashboard.server --mode live --rover-url http://192.168.4.1/js --host 127.0.0.1 --p
```

Optional comparison policies

```
python -m DigitalTwin.dashboard.server --mode live --rover-url http://192.168.4.1/js --host 127.0.0.1 --p
python -m DigitalTwin.dashboard.server --mode live --rover-url http://192.168.4.1/js --host 127.0.0.1 --p
python -m DigitalTwin.dashboard.server --mode live --rover-url http://192.168.4.1/js --host 127.0.0.1 --p
```

Step 3: Drive one controlled live session

- 5 s stationary start
- straight forward
- straight reverse
- one or two gentle turns
- a short turning-intensive segment
- 5 s stationary end

Step 4: Save the live log

The dashboard automatically writes JSONL logs to `raw_logs/live_validation/`.

```
Get-ChildItem .\raw_logs\live_validation | Sort-Object LastWriteTime -Descending | Select-Object -First 1
```

5. What The Live Experiment Produces

- dashboard visualization
- screenshots or screen recording
- live contract decisions
- live policy decisions
- JSONL session logs

It does not by itself produce final AprilTag-based ATE, RPE, or heading metrics.

6. When AprilTags Are Needed

Use AprilTags only if you want independent physical-validation metrics: ATE, RPE, heading error, and physical-versus-virtual trajectory comparison. That is a separate paired experiment: collect telemetry CSV, record overhead video, then run the offline AprilTag analysis pipeline.

7. Should GPS Stay Connected?

Yes. For the live contract experiment, GPS should stay connected.

- the dashboard uses live GPS-versus-twin agreement
- it makes the contract panels meaningful in real time
- the later follow-up work still needs GPS

If GPS is disconnected, the twin can still propagate from IMU and encoders, but the live contract evidence becomes much less persuasive.

8. Important Limitation

Do not run multiple simultaneous aggressive telemetry pollers against the rover unless you intentionally want extra network load. Safest workflow: run the live dashboard experiment by itself, then run any offline telemetry/AprilTag validation as a separate trial.

9. Minimal Checklist

- 1 Keep **GPS connected**.
- 2 Do not worry about AprilTags for the live demo.
- 3 Power the rover and confirm `http://192.168.4.1`.
- 4 Run the live dashboard command.
- 5 Drive one controlled session.
- 6 Save screenshots and the JSONL log.
- 7 If you later want physical fidelity metrics, run a separate AprilTag session.

10. Related Repo Files

- `docs/ugv01_live_validation_runbook.md`
- `docs/ugv01_live_validation_dashboard.md`
- `docs/ugv01_esp32_bringup.md`
- `DigitalTwin/configs/ugv01_live_service_contracts.json`

Practical rule: for the live contract experiment, use GPS and the dashboard only. Use AprilTags later when you want independent offline metrics.